25901281 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFSC200 - FORENSIC SEROLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Define antigen–antibody reaction. [U] / [CO1]
2. Justify the use of cross-over electrophoresis as a confirmatory test [E] / [CO4]
3. Name one confirmatory test for blood. [K] / [CO2]
4. Define blood grouping from dried blood stains. [K] / [CO4]
5. Explain the significance of blood group determination in forensic investigation. [U] / [CO3]
6. Define rouleaux formation. [U] / [CO2]
7. Demonstrate the step-by-step procedure of the Kastle–Mayer test. [S] / [CO4]
8. Why are serological tests useful in identification of biological fluids? [U] / [CO1]
9. Apply Esterase D typing in personal identification. [U] / [CO1]
10. Describe laboratory detection of iso-enzymes. [U] / [CO1]
11. Explain forensic importance of polymorphic proteins. [K] / [CO1]

12. Which test would you apply to screen a suspected bloodstain at the scene? [A] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. Analyse the role of bloodstain pattern analysis in crime scene reconstruction. [An] / [CO4]

14. Write microscopic examination to confirm the presence of spermatozoa. [A] / [CO2]

15. Describe collection and preservation of various body fluids. [A] / [CO4]

16. Define affinity and avidity in antigen–antibody reactions. [U] / [CO1]

17. Explain the procedure for blood group determination from fresh blood. [A] / [CO2]

18. Justify the need for proper handling and preservation of semen samples. [E] / [CO1]

19. Describe the theory of inheritance of ABO blood groups with a suitable genetic explanation. [U] / [CO2]

[4x5 = 20]

Part C

Essay questions

Answer any **1** questions

Each question carries **10** marks

20. Apply HLA typing in criminal and paternity investigations. [A] / [CO1]

21. Explain the principles and procedures of presumptive and confirmatory tests for blood. Discuss the Kastle–Mayer (Phenolphthalein) test and Takayama test in detail. [K] / [CO2]

[10x1 = 10]